Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-17 SWL Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.352 Hz*
Freq. upper bound:	*7.031 Hz*

Freq. Domain Analysis

Overlap of consecutive series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*27*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *27.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1274*	*0.03488*
Peak wave period T_P (s):	*1.422*	*1.455*
Energy wave period $T_{-1,0}$ (s):	*1.345*	*1.077*
Mean wave period $T_{0,1}$ (s):	*1.26*	*0.8425*
Mean wave period $T_{0,2}$ (s):	*1.199*	*0.746*
Spectral width (Broadness):	*0.7215*	*0.7722*
Spectral width (Narrowness):	*0.3222*	*0.5248*

Time Domain Analysis

Overall amplitude reflection coefficient: *27.4%*

	Incident	**Reflected**
Number of waves:	1019	1501
Mean wave height H_m (m):	*0.08386*	*0.02201*
Mean wave period T_m (s):	*1.286*	*0.8732*
Sig. wave height H_s (m):	*0.1251*	*0.03415*
$T_{H_{1/3}}$ (s):	*1.402*	*1.044*
H_{max} (m):	*0.1953*	*0.06438*
T_{Hmax} (s):	*1.575*	*0.5827*
$H_{1/10}$ (m):	*0.1507*	*0.04351*
$H_{1/50}$ (m):	*0.1724*	*0.05437*
$H_{1/100}$ (m)	*0.1791*	*0.05751*
$H_{1/250}$ (m):	*0.1895*	*0.06039*
$H_{0.1\%}$ (m):	*0.1953*	*0.06254*
$H_{1\%}$ (m):	*0.1703*	*0.05429*
$H_{2\%}$ (m):	*0.163*	*0.04821*
$H_{10\%}$ (m)	*0.1347*	*0.03635*
Groupiness factor GF:	*0.9093*	*1.147*
Skewness b1:	*0.3906*	*0.2384*
Kurtosis b2:	*2.797*	*3.5*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-17 SWL Wide Berm 4% - Test 1.experiment_1.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.336 Hz*
Freq. upper bound:	*6.719 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *28.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1381*	*0.03974*
Peak wave period T_P (s):	*1.488*	*1.488*
Energy wave period $T_{-1,0}$ (s):	*1.398*	*1.116*
Mean wave period $T_{0,1}$ (s):	*1.296*	*0.8453*
Mean wave period $T_{0,2}$ (s):	*1.222*	*0.7393*
Spectral width (Broadness):	*0.762*	*0.7837*
Spectral width (Narrowness):	*0.3541*	*0.5544*

Time Domain Analysis

Overall amplitude reflection coefficient: *28.8%*

	Incident	**Reflected**
Number of waves:	1020	1512
Mean wave height H_m (m):	*0.09142*	*0.02546*
Mean wave period T_m (s):	*1.34*	*0.9043*
Sig. wave height H_s (m):	*0.136*	*0.03917*
$T_{H_{1/3}}$ (s):	*1.47*	*1.075*
H_{max} (m):	*0.1958*	*0.07948*
T_{Hmax} (s):	*1.61*	*0.6074*
$H_{1/10}$ (m):	*0.1581*	*0.04923*
$H_{1/50}$ (m):	*0.1786*	*0.06003*
$H_{1/100}$ (m)	*0.1836*	*0.06408*
$H_{1/250}$ (m):	*0.1907*	*0.06928*
$H_{0.1\%}$ (m):	*0.1958*	*0.07516*
$H_{1\%}$ (m):	*0.175*	*0.05928*
$H_{2\%}$ (m):	*0.1711*	*0.05397*
$H_{10\%}$ (m)	*0.1435*	*0.0421*
Groupiness factor GF:	*0.8945*	*1.116*
Skewness b1:	*0.4774*	*0.1583*
Kurtosis b2:	*2.77*	*3.395*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-17 SWL Wide Berm 4% - Test 1.experiment_4.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.320 Hz*
Freq. upper bound:	*6.406 Hz*

Freq. Domain Analysis

Overlap of (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*30*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *29.2%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1446*	*0.04228*
Peak wave period T_P (s):	*1.561*	*1.561*
Energy wave period $T_{-1,0}$ (s):	*1.45*	*1.164*
Mean wave period $T_{0,1}$ (s):	*1.332*	*0.8592*
Mean wave period $T_{0,2}$ (s):	*1.244*	*0.7417*
Spectral width (Broadness):	*0.7932*	*0.7991*
Spectral width (Narrowness):	*0.3815*	*0.5848*

Time Domain Analysis

Overall amplitude reflection coefficient: *29.2%*

	Incident	**Reflected**
Number of waves:	1034	1521
Mean wave height H_m (m):	*0.09602*	*0.02754*
Mean wave period T_m (s):	*1.38*	*0.9387*
Sig. wave height H_s (m):	*0.1428*	*0.0423*
$T_{H_{1/3}}$ (s):	*1.526*	*1.134*
H_{max} (m):	*0.214*	*0.08445*
T_{Hmax} (s):	*1.519*	*0.8706*
$H_{1/10}$ (m):	*0.166*	*0.05277*
$H_{1/50}$ (m):	*0.1867*	*0.06323*
$H_{1/100}$ (m)	*0.1949*	*0.0676*
$H_{1/250}$ (m):	*0.2038*	*0.07333*
$H_{0.1\%}$ (m):	*0.2138*	*0.08115*
$H_{1\%}$ (m):	*0.1854*	*0.06173*
$H_{2\%}$ (m):	*0.1756*	*0.05646*
$H_{10\%}$ (m)	*0.1502*	*0.04554*
Groupiness factor GF:	*0.8783*	*1.092*
Skewness b1:	*0.5145*	*0.1633*
Kurtosis b2:	*2.737*	*3.306*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-17 SWL Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.313 Hz*
Freq. upper bound:	*6.250 Hz*

Freq. Domain Analysis

Overlapping of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*28*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*31*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *30.2%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1582*	*0.04771*
Peak wave period T_P (s):	*1.6*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.507*	*1.218*
Mean wave period $T_{0,1}$ (s):	*1.373*	*0.8799*
Mean wave period $T_{0,2}$ (s):	*1.276*	*0.7553*
Spectral width (Broadness):	*0.8026*	*0.7995*
Spectral width (Narrowness):	*0.3977*	*0.5978*

Time Domain Analysis

Overall amplitude reflection coefficient: *30.2%*

	Incident	**Reflected**
Number of waves:	1032	1529
Mean wave height H_m (m):	*0.1061*	*0.03123*
Mean wave period T_m (s):	*1.441*	*0.9729*
Sig. wave height H_s (m):	*0.1535*	*0.04754*
$T_{H_{1/3}}$ (s):	*1.6*	*1.22*
H_{max} (m):	*0.2163*	*0.08115*
T_{Hmax} (s):	*1.452*	*0.66*
$H_{1/10}$ (m):	*0.1765*	*0.05853*
$H_{1/50}$ (m):	*0.1981*	*0.06913*
$H_{1/100}$ (m)	*0.2048*	*0.07253*
$H_{1/250}$ (m):	*0.2103*	*0.07623*
$H_{0.1\%}$ (m):	*0.2161*	*0.08092*
$H_{1\%}$ (m):	*0.1983*	*0.06762*
$H_{2\%}$ (m):	*0.1875*	*0.06364*
$H_{10\%}$ (m)	*0.161*	*0.05053*
Groupiness factor GF:	*0.8463*	*1.068*
Skewness b1:	*0.5507*	*0.1758*
Kurtosis b2:	*2.657*	*3.209*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-17 SWL Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.297 Hz*
Freq. upper bound:	*5.938 Hz*

Freq. Domain Analysis

Overlap of consecutive series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*30*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*32*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *30.6%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1619*	*0.0495*
Peak wave period T_P (s):	*1.641*	*1.73*
Energy wave period $T_{-1,0}$ (s):	*1.536*	*1.266*
Mean wave period $T_{0,1}$ (s):	*1.385*	*0.8908*
Mean wave period $T_{0,2}$ (s):	*1.278*	*0.7563*
Spectral width (Broadness):	*0.8143*	*0.8053*
Spectral width (Narrowness):	*0.418*	*0.6223*

Time Domain Analysis

Overall amplitude reflection coefficient: *30.7%*

	Incident	**Reflected**
Number of waves:	1060	1514
Mean wave height H_m (m):	*0.1077*	*0.0327*
Mean wave period T_m (s):	*1.463*	*1.024*
Sig. wave height H_s (m):	*0.1572*	*0.04938*
$T_{H_{1/3}}$ (s):	*1.639*	*1.264*
H_{max} (m):	*0.2267*	*0.09242*
T_{Hmax} (s):	*1.466*	*0.5972*
$H_{1/10}$ (m):	*0.1821*	*0.0604*
$H_{1/50}$ (m):	*0.2054*	*0.07404*
$H_{1/100}$ (m)	*0.2115*	*0.07843*
$H_{1/250}$ (m):	*0.2201*	*0.08342*
$H_{0.1\%}$ (m):	*0.2265*	*0.089*
$H_{1\%}$ (m):	*0.2035*	*0.0722*
$H_{2\%}$ (m):	*0.1951*	*0.06658*
$H_{10\%}$ (m)	*0.1665*	*0.05229*
Groupiness factor GF:	*0.8637*	*1.074*
Skewness b1:	*0.6061*	*0.195*
Kurtosis b2:	*2.716*	*3.238*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-17 SWL Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.273 Hz*
Freq. upper bound:	*5.469 Hz*

Freq. Domain Analysis

Overlapping of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*30*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*35*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *31.1%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1644*	*0.05113*
Peak wave period T_P (s):	*1.778*	*1.829*
Energy wave period $T_{-1,0}$ (s):	*1.575*	*1.334*
Mean wave period $T_{0,1}$ (s):	*1.409*	*0.9111*
Mean wave period $T_{0,2}$ (s):	*1.295*	*0.7666*
Spectral width (Broadness):	*0.8141*	*0.8056*
Spectral width (Narrowness):	*0.4295*	*0.6423*

Time Domain Analysis

Overall amplitude reflection coefficient: *31.2%*

	Incident	**Reflected**
Number of waves:	1022	1416
Mean wave height H_m (m):	*0.1107*	*0.03423*
Mean wave period T_m (s):	*1.516*	*1.096*
Sig. wave height H_s (m):	*0.1608*	*0.0517*
$T_{H_{1/3}}$ (s):	*1.685*	*1.33*
H_{max} (m):	*0.2266*	*0.09553*
T_{Hmax} (s):	*1.875*	*1.396*
$H_{1/10}$ (m):	*0.1874*	*0.06389*
$H_{1/50}$ (m):	*0.2086*	*0.07736*
$H_{1/100}$ (m)	*0.2156*	*0.08357*
$H_{1/250}$ (m):	*0.2224*	*0.09149*
$H_{0.1\%}$ (m):	*0.2266*	*0.09516*
$H_{1\%}$ (m):	*0.2087*	*0.07336*
$H_{2\%}$ (m):	*0.1966*	*0.06921*
$H_{10\%}$ (m)	*0.1715*	*0.05518*
Groupiness factor GF:	*0.8791*	*1.099*
Skewness b1:	*0.6207*	*0.1971*
Kurtosis b2:	*2.76*	*3.247*

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Reflection Analysis

Input Parameters

Generel

Data file:	*ROR 2026-3-17 SWL Wide Berm 4% - Test 1 Overall - Subseries (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of subblocks:	*60%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *25%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.08675*	*0.02171*
Peak wave period T_P (s):	*1.524*	*1.6*
Energy wave period $T_{-1,0}$ (s):	*1.444*	*1.353*
Mean wave period $T_{0,1}$ (s):	*1.366*	*1.119*
Mean wave period $T_{0,2}$ (s):	*1.315*	*0.9879*
Spectral width (Broadness):	*0.602*	*0.788*
Spectral width (Narrowness):	*0.2806*	*0.5312*

Time Domain Analysis

Overall amplitude reflection coefficient: *25.2%*

	Incident	**Reflected**
Number of waves:	999	1253
Mean wave height H_m (m):	*0.05521*	*0.01313*
Mean wave period T_m (s):	*1.369*	*1.091*
Sig. wave height H_s (m):	*0.08457*	*0.02177*
$T_{H_{1/3}}$ (s):	*1.473*	*1.388*
H_{max} (m):	*0.1404*	*0.04173*
T_{Hmax} (s):	*1.64*	*0.6982*
$H_{1/10}$ (m):	*0.1033*	*0.02853*
$H_{1/50}$ (m):	*0.1209*	*0.03491*
$H_{1/100}$ (m)	*0.1301*	*0.03693*
$H_{1/250}$ (m):	*0.139*	*0.03965*
$H_{0.1\%}$ (m):	*Not enough data*	*0.04141*
$H_{1\%}$ (m):	*0.1215*	*0.034*
$H_{2\%}$ (m):	*0.1088*	*0.03213*
$H_{10\%}$ (m)	*0.09134*	*0.02392*
Groupiness factor GF:	*0.9673*	*1.181*
Skewness b1:	*0.3443*	*0.1697*
Kurtosis b2:	*2.919*	*3.609*

Reflection Analysis

Input Parameters

Generel

Data file:	*ROR 2026-3-17 SWL Wide Berm 4% - Test 1 Experiment subseries (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.266 Hz*
Freq. upper bound:	*5.313 Hz*

Freq. Domain Analysis

Overlap of subblocks:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*29*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*36*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *29.3%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1074*	*0.03144*
Peak wave period T_P (s):	*1.73*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.618*	*1.579*
Mean wave period $T_{0,1}$ (s):	*1.491*	*1.217*
Mean wave period $T_{0,2}$ (s):	*1.406*	*1.026*
Spectral width (Broadness):	*0.7098*	*0.833*
Spectral width (Narrowness):	*0.3522*	*0.6371*

Time Domain Analysis

Overall amplitude reflection coefficient: *29.3%*

	Incident	**Reflected**
Number of waves:	1003	1137
Mean wave height H_m (m):	*0.06831*	*0.02001*
Mean wave period T_m (s):	*1.521*	*1.343*
Sig. wave height H_s (m):	*0.1063*	*0.03286*
$T_{H_{1/3}}$ (s):	*1.696*	*1.656*
H_{max} (m):	*0.1872*	*0.06658*
T_{Hmax} (s):	*1.658*	*1.33*
$H_{1/10}$ (m):	*0.1326*	*0.0431*
$H_{1/50}$ (m):	*0.1621*	*0.05313*
$H_{1/100}$ (m)	*0.1742*	*0.05662*
$H_{1/250}$ (m):	*0.1842*	*0.06112*
$H_{0.1\%}$ (m):	*0.1872*	*0.0658*
$H_{1\%}$ (m):	*0.1565*	*0.05137*
$H_{2\%}$ (m):	*0.147*	*0.04786*
$H_{10\%}$ (m)	*0.1132*	*0.03496*
Groupiness factor GF:	*1.076*	*1.26*
Skewness b1:	*0.5629*	*0.258*
Kurtosis b2:	*3.298*	*3.629*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*23. mrt 2026*
Remarks	
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